Week 18 — Extended Reading: Preference models and PPO for language

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Christiano et al. (2017): Human preferences

Primary source. https://arxiv.org/abs/1706.03741

Mechanism and scope. The paper learns a reward signal from pairwise trajectory preferences and uses it for policy learning. It makes preference collection and reward generalisation part of the feedback loop.

Read, derive, investigate. Why is a comparison probability a model of noisy preference rather than a deterministic ordering rule? Add the same constant to both scores and show the likelihood is unchanged. Use this invariance to discuss what a fitted score does and does not identify.

2 Schulman et al. (2017): PPO

Primary source. https://arxiv.org/abs/1707.06347

Mechanism and scope. PPO proposes practical surrogate objectives for policy updates, including clipped probability ratios. The clipped objective limits the incentive to move selected sampled terms further; it does not constrain every parameter update or impose a hard KL budget.

Read, derive, investigate. Plot a single positive-advantage and negative-advantage term against the probability ratio. Which side flattens? Distinguish the old rollout policy from the fixed language reference policy. They support different ratios and KL terms.

3 Ziegler et al. (2019): Language fine-tuning

Primary source. https://arxiv.org/abs/1909.08593

Mechanism and scope. This work adapts language models from human preferences with a reference-policy KL penalty and an adaptive coefficient. The coefficient responds to measured divergence; controller cadence and whether KL is per token or per sequence affect the update.

Read, derive, investigate. Write the sign of the coefficient update when observed KL exceeds its target. Check the units of sample count and controller horizon. The workbook uses one controller step at a time; it does not silently reproduce every scaling convention in the original training system.

4 Ouyang et al. (2022): InstructGPT

Primary source. https://arxiv.org/abs/2203.02155

Mechanism and scope. InstructGPT combines supervised demonstrations, preference-based reward modelling and policy optimisation. The paper's evaluation compares particular models and feedback protocols, not all possible human values or all future use cases.

Read, derive, investigate. Draw three separate datasets and explain which loss each trains. Where can annotator preferences, prompt selection or reward-model error affect the final policy? The local sentiment experiment demonstrates the pipeline's mechanics, not the paper's model scale or evaluation.

5 Gao et al. (2022): Reward overoptimisation

Primary source. https://arxiv.org/abs/2210.10760

Mechanism and scope. This empirical study compares optimisation of a proxy reward with evaluation by a stronger reward model. Its fitted scaling relationships describe studied regimes, rather than universal analytic laws of all reward misspecification.

Read, derive, investigate. Put proxy score and evaluation score on the same optimisation-budget axis. What observation would falsify the claim that more optimisation always helps? Explain why a held-out reward model is itself still a proxy rather than an infallible human-quality oracle.

6 Bai et al. (2022): Constitutional AI

Primary source. https://arxiv.org/abs/2212.08073

Mechanism and scope. Constitutional AI uses principles in critique/revision and AI-generated preference feedback. Human choices remain in principles, model construction and evaluation; replacing some labels does not remove those normative inputs.

Read, derive, investigate. Trace how a principle becomes a training signal. Design a response that follows a superficial wording cue while violating the principle's intent. Compare held-out human assessment against agreement with the same model used to generate labels.

7 Stiennon et al. (2020): Summarisation

Primary source. https://arxiv.org/abs/2009.01325

Mechanism and scope. The work studies summarisation trained with human feedback and reports evaluations beyond reward-model fit. It is a concrete setting for separating preference prediction accuracy from policy quality.

Read, derive, investigate. What can happen if summaries grow more persuasive but less faithful? Name an evaluation that checks source-grounded factuality. A reward-model win rate alone does not establish that property.

8 Lambert et al. (2024): Tulu 3

Primary source. https://arxiv.org/abs/2411.15124

Mechanism and scope. Tulu 3 documents a multi-stage open post-training recipe, including supervised tuning, preference optimisation and reinforcement learning with verifiable rewards. Stages and datasets are explicit parts of its experimental design.

Read, derive, investigate. Separate the stage order from an arbitrary mixture of loss terms. Which changes would require a new ablation? The repository's optional LLM lab supplies SFT/DPO smoke workflows; the week's NumPy/PyTorch helpers are not a current universal TRL trainer API.

Connect the reading to your workbook

Before running a policy update, hand-calculate a two-comparison Bradley–Terry loss, the per-token reference log ratio, and the final-token reward. Verify terminal value masking in GAE. Record the preference model, reference model and rollout-policy snapshots separately so their roles cannot collapse into one variable.

Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.

Statistics reading bridge

Twenty preference pairs may contain only two prompt units. Read dependence and empirical resampling before presenting preference accuracy intervals. Report prompt and pair counts separately, preserve prompt clusters, and retain RLHF primary references for reward modeling and KL-regularized policy optimization.

Checked textbook locations: Siegrist § § 4.5, 6.2, PDF pp.478, 817; Ross § 15.3, PDF p.683. https://stats.libretexts.org/@go/page/10160. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.